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STAFF & ADMINISTRATOR GUIDE

WaitLess

Digital Hospital Queue Management System

Chinhoyi Provincial Hospital

Complete operational manual for all staff roles.

Version 1.0 | August 2026

What's Inside

Login & Security • Reception • Triage • Clinician Workflow

Pharmacy • Laboratory • OI Clinic • Admin • Reports

Notifications • Display Boards • Troubleshooting

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1. System Overview

WaitLess is a digital queue management system that replaces paper tickets with real-time digital tracking. Patients register, get triaged, wait in a virtual queue, receive WhatsApp notifications, and are called by staff — all managed through a web-based interface accessible on desktops, tablets, and phones.

The Patient Journey

```
—— PATIENT FLOW ——

Registration → Triage Assessment → Queue Waiting → Called by Clinician
→ Consultation → [Transfer to Lab/Pharmacy/Radiology if needed]
→ Completion → Feedback
```

Key Components

- **Backend server** — Node.js, runs on hospital LAN (port 4242)
- **Frontend** — React web app, works in any modern browser
- **Database** — MySQL for production, file-based JSON for development
- **Real-time** — Server-Sent Events (SSE) push live updates to all screens
- **Notifications** — WhatsApp Cloud API with automatic retry queue
- **AI Service** — Wait time prediction, bottleneck analysis, patient chatbot
- **Audit Trail** — SHA-256 hash-chained, tamper-evident action log

Staff Roles

Administrator

Full access. Manages staff, configuration, hospitals, audit trail, sync/backup.

Receptionist

Patient registration, search, ticket generation, kiosk management.

Nurse (Triage)

Triage assessment, vital signs, priority assignment, reassessment alerts.

Doctor (Clinician)

Call next patient, consultations, transfers, test orders, prescriptions, completion.

Pharmacist

Lab Technician

Pharmacy queue, prescription dispensing, stock flags, ready/collected states.

Lab queue, test order lifecycle, turnaround tracking, result-ready notifications.

Radiologist

Radiology queue, imaging orders, result routing back to clinician.

All Staff

Dashboard access, queue visibility, notification panel, profile management.

2. Logging In & Security

Logging In

1 **Open the staff login page**

Navigate to `/admin/login` in your browser. Enter your username or email and password.

2 **You are redirected to your role's landing page**

Receptionists go to registration, triage nurses to triage, clinicians to dashboard, admins to admin dashboard.

Password Requirements

- Minimum 10 characters
- At least 3 of: uppercase letters, lowercase letters, numbers, symbols
- Cannot be a commonly used password
- Passwords are hashed with bcrypt (12 rounds) — never stored in plain text

Security notes:

- Your session automatically logs out after 8 hours of inactivity
- Never share your login credentials with another staff member
- All actions are recorded in the audit trail with your identity
- Only administrators can create new staff accounts

Forgot Password?

Click "Forgot password?" on the login page. Enter your email address. A reset link will be generated. See your administrator if you need help resetting your password.

3. For Reception Staff

Registering a New Patient

1 **Open the registration page**

Click "Register" in the sidebar navigation. The patient registration form appears.

2 **Search for returning patients first**

Use the search bar to look up existing patients by name, phone, national ID, patient number, or scan their QR code. If found, reuse their profile — avoid creating duplicates.

3 **Capture patient details for new patients**

Fill in all required fields:

- Full name, national ID, date of birth, gender
- Phone number (for WhatsApp notifications)
- Address
- Patient category: Walk-in, Referred, Emergency, Elderly, Pregnant, Child, Disabled, or Chronic
- Next of kin name and phone number
- Chief complaint (what's wrong)
- Target department: OPD, Casualty, Pharmacy, Lab, Radiology, or OI Clinic
- Notification consent checkbox

4 **Submit and print ticket**

Click "Register Patient". A ticket is generated (e.g., **OPD-205**) with a QR code. Print the 80mm ticket or share to the patient's phone. The patient appears in the queue immediately.

> **Form auto-save:** If your browser crashes or you accidentally close the tab, the form data is automatically saved. When you return, your entries will be restored.

Managing Patient Search

- Search by any identifier: name, phone, national ID, patient number, or QR code
- Click on a patient to see their visit history and current tickets
- Always search before creating a new record to avoid duplicates

4. For Triage Nurses

Performing a Triage Assessment

1 Open the triage console

Click "Triage" in the sidebar. You see all waiting patients sorted by arrival time.

2 Select a patient

Click on a patient to open the triage assessment form.

3 Capture clinical data

Record:

- **Chief complaint** — what the patient reports
- **Vital signs:** Blood pressure, pulse, temperature, respiratory rate, oxygen saturation
- **Pain level:** 0–10 scale
- **Symptoms and severity**
- **Emergency indicators** — system auto-detects critical thresholds

4 Assign priority

Choose one:

- **RED** — Resuscitation (immediate, life-threatening). Patient jumps to front of queue.
- **YELLOW** — Urgent (should be seen within 60 minutes)
- **GREEN** — Standard (routine queue order)
- **BLACK** — Deceased/expectant

The queue automatically re-sorts by priority. RED patients go to the front immediately.

Responding to Reassessment Alerts

> **Critical safety feature:** If a RED or YELLOW patient has been waiting longer than the safe threshold, a reassessment alert appears on the staff dashboard. Click the alert to re-evaluate the patient's condition. You

> can upgrade their priority if their condition has worsened.

5. For Clinicians

Calling the Next Patient

1 **Open the staff dashboard**

Navigate to `/admin/dashboard`. You see the live queue for your department with priority colors and wait times.

2 **Click "Call Next"**

The system calls the highest-priority, longest-waiting patient. The patient's ticket appears on the public board as "Now Serving".

3 **Patient is notified**

A WhatsApp "called" notification is sent. Audio announcement plays (if enabled). The patient proceeds to your room.

After Consultation — Your Options

Action	What happens
Mark Complete	Patient visit is finished. Patient can now give feedback via portal.
Transfer to Department	Patient moves to Lab, Pharmacy, Radiology, or OI Clinic queue. Gets new ticket number and WhatsApp notification.
Order Tests	Create lab or radiology orders with test type and priority. Orders appear in the Lab/Radiology module.
Prescribe	Route prescription to pharmacy. Patient appears in Pharmacy queue.

Managing Missed Patients

1

Mark as missed

If a patient doesn't respond when called, click "Mark as Missed". The patient moves to the "Missed" lane on the board. A missed-turn WhatsApp notification is sent.

2

Recall when patient returns

When the patient returns, click "Recall" to reinsert them into the queue fairly — they don't go to the back of the line.

6. For Pharmacists

Pharmacy Workflow

1 **Open the Pharmacy module**

Navigate to `/admin/pharmacy`. See all patients in the pharmacy queue with prescription status.

2 **Check prescription details**

Click on a patient to see their prescription items, dosages, and quantities.

3 **Check stock availability**

If an item is out of stock, flag it. The patient is automatically notified. The out-of-stock event is recorded for reports.

4 **Dispense and mark ready**

Once medication is dispensed, mark the prescription as "Ready". The patient receives a WhatsApp notification: "Your prescription is ready at the Pharmacy window."

5 **Mark as collected**

When the patient picks up their medication, mark as "Collected" to complete the pharmacy visit.

7. For Lab Technicians

Laboratory Workflow

1 **Open the Laboratory module**

Navigate to `/admin/laboratory`. See all pending test orders with patient and test details.

2 **Update test lifecycle**

Track each test through its lifecycle:

- **Pending** — Order received from clinician
- **Sample Collected** — Specimen obtained from patient
- **In Progress** — Test is running
- **Completed / Result Ready** — Results available

3 **Notify when results are ready**

When you mark a test as "Result Ready", the system automatically:

- Sends a WhatsApp notification to the patient
- Routes the patient back to the doctor's queue for follow-up

> **Turnaround time tracking:** The system records the time at each lifecycle stage. Lab turnaround reports show average time from order to result, helping identify slow tests.

8. For OI Clinic Staff

OI Clinic Workflow

The OI (Opportunistic Infection) Clinic handles sensitive services including HIV/TB care. Patient privacy is protected through confidential handling.

Service Streams

Stream	Description
ART	Antiretroviral Therapy management and refills
TB	Tuberculosis screening and treatment
Viral Load	Viral load monitoring and testing
Counselling	Patient counselling sessions
Cervical Screening	Cervical cancer screening

> **Confidentiality:** OI Clinic patients have masked routing on public display boards. Their personal details are only visible to authorized OI Clinic staff.

Managing OI Visits

- View patients by service stream
- Create new OI visits linked to patient tickets
- Update visit status and outcomes
- View OI summary dashboard for the day

9. For Administrators

Creating Staff Accounts

1 **Go to staff registration**

Navigate to `/admin/staff-register`. This page is only accessible to administrators.

2 **Fill in staff details**

Enter: full name, username, email, phone, employee ID, department, role, password (10+ chars with 3 of 4 character types), and optionally a profile photo.

3 **Submit**

Click "Create staff account". The new staff member can now log in with their credentials. The creation is recorded in the audit trail.

Viewing the Audit Trail

- Navigate to `/admin/audit`
- See every action taken in the system: who did it, when, what changed
- The audit log is SHA-256 hash-chained — any tampering is mathematically detectable
- Filter by action type, staff member, or date range

System Configuration

- Navigate to `/admin/settings`
- Configure hospital information (name, location, contact)
- Manage departments and their settings
- Configure notification preferences
- Reset configuration to defaults if needed

Multi-Hospital Management

- Navigate to the hospitals management section
- View, create, and update hospital profiles

- Each hospital has its own configuration and departments

Backup and Sync

- **Export** — Download all system data as a JSON backup
- **Import** — Restore from a previous backup file
- **Backup** — Create a server-side backup
- **Status** — Check sync status and data integrity

10. Reports & Analytics

WaitLess generates 10 types of operational reports, all exportable to CSV, Excel (XLSX), and PDF.

Report	What it shows
Average Wait Time	Average waiting time per department with historical trends
Department Performance	Patients served, service time, queue length per department
No-Show Report	Missed turns correlated with notification delivery status
Triage Report	Triage assessments with timing, outcomes, and override tracking
Lab Turnaround	Time from test order to result ready, per test type
Staff Workload	Patients handled per staff member, service time, productivity
Out-of-Stock	Pharmacy stock flags with frequency and affected patients
Peak Hours	Patient arrival patterns by hour and day for resource planning
Congestion Report	AI-powered queue congestion prediction per department
Resource Allocation	AI-powered suggestions for staff deployment

Generating a Report

1 **Go to Reports**

Navigate to `/admin/reports`. Select the report type from the sidebar.

2 **Select date range**

Choose the start and end dates for the report period.

3 **View and export**

Results appear in tables and charts. Click "Export CSV", "Export Excel", or "Export PDF" to download.

AI Analytics

- **Wait Time Prediction** — Predicts how long new patients will wait based on historical data
- **Bottleneck Analysis** — Identifies which departments are congested and recommends actions
- **Resource Allocation** — Suggests where to deploy staff for maximum impact

11. Notifications & Display Boards

WhatsApp Notifications

The system sends WhatsApp messages automatically at key points in the patient journey. You can monitor delivery status on the dashboard.

Trigger	Message sent
Patient registered	Confirmation with ticket code and tracking URL
Near turn (2-3 ahead)	"Your turn is approaching. Please return to [department]."
Patient called	"[Ticket], please proceed to [room]."
Missed turn	"You missed your turn. Please see reception."
Transferred	"You have been moved to [department]. New ticket: [code]."
Prescription ready	"Your prescription is ready at the Pharmacy window."
Lab results ready	"Your lab results are ready. Please return to the doctor."

Notification Retry

> **Automatic retry:** If a WhatsApp message fails to send (e.g., internet outage), it enters a retry queue. The system retries at 15 seconds, 60 seconds, and 5 minutes. You can also manually retry from the dashboard notification panel.

Display Boards

Screen	URL	Where to display
Public Queue Board	`/queue`	Large TV in main waiting area
Department Board	`/department-board`	TV at each department's waiting area

> **Setup tip:** Set display board browsers to kiosk mode (full screen, no address bar) for a clean public display. The boards update in real time via Server-Sent Events — no page refresh needed.

12. Appointments

Viewing Appointments

- Navigate to `/admin/appointments`
- View all appointments across departments
- Filter by date, department, or status
- See appointment summary statistics

Managing Appointments

- View appointment details and patient information
- Update appointment status (confirmed, cancelled, completed, no-show)
- View available slots per date and service category
- Track appointment reminders sent to patients

Follow-Up Appointments

- Create follow-up appointments linked to completed visits
- View all follow-ups for a specific patient ticket
- Track reminder delivery for follow-up appointments

13. Troubleshooting

"Unable to connect to the server"

The backend server is not running or is unreachable. Check:

- Is the backend server process running? (Should be on port 4242)
- Is the hospital LAN connected?
- Is the correct API URL configured in the frontend?

WhatsApp messages not being sent

- Check WhatsApp status at `/admin/notifications`
- Verify WHATSAPP_PROVIDER is set to "cloud" (not "mock") in backend .env
- Ensure WHATSAPP_ACCESS_TOKEN and WHATSAPP_PHONE_NUMBER_ID are set
- Check that the patient has WhatsApp consent and a valid phone number
- Messages in the retry queue will send automatically when connectivity returns

Staff account creation fails

- Password must be 10+ characters with at least 3 of: uppercase, lowercase, number, symbol
- Username must be 3-32 characters, only letters, numbers, dots, underscores, hyphens
- Check for duplicate username, email, or employee ID
- Only administrators can create staff accounts

Display board not updating

- Check that the backend server is running (SSE events stream from /api/events)
- Refresh the display board page
- Check browser console for EventSource connection errors

Patient not appearing in queue

- Verify the patient was registered to the correct department
- Check the queue board filter settings
- Ensure the patient's status is "waiting" (not "completed" or "missed")

Form data lost after browser crash

WaitLess auto-saves form drafts. When you return to the registration or triage page, your previous entries should be restored. If not, the auto-save may have been interrupted — re-enter the data.

AI chatbot not responding

- The chatbot requires the backend server to be running
- Check that the /api/ai/chatbot endpoint is accessible
- The chatbot uses keyword matching — try rephrasing your question
- Topics supported: wait times, ticket tracking, prescriptions, lab tests, appointments, emergency guidance

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Built for Chinhoyi Provincial Hospital • Version 1.0 • August 2026
Replacing confusion with a queue patients can trust.